

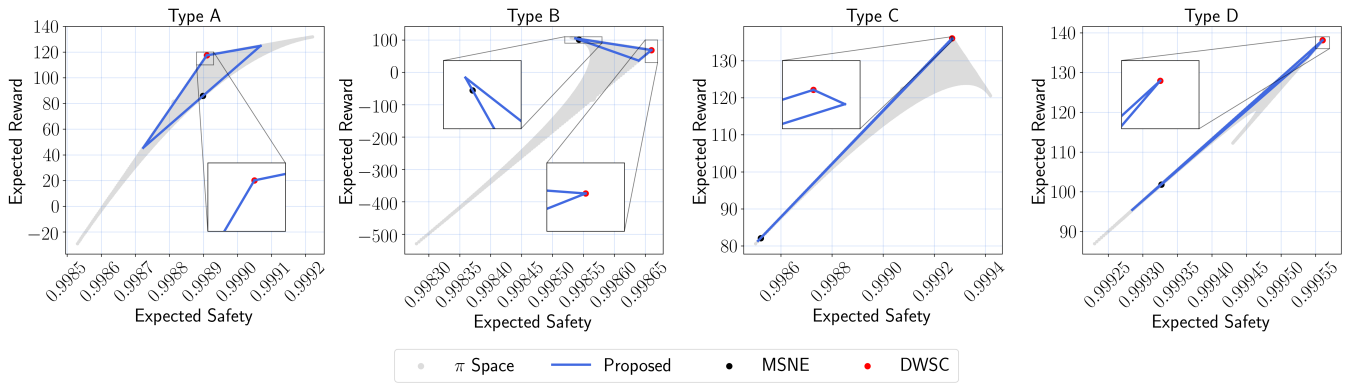


Figure 5: Illustration of achievable rewards and safety of different methods. The gray area labeled as ‘ π Space’ denotes all possible interaction strategies. Note that the outcome of DWSC and MSNE methods are always achievable by the proposed method.

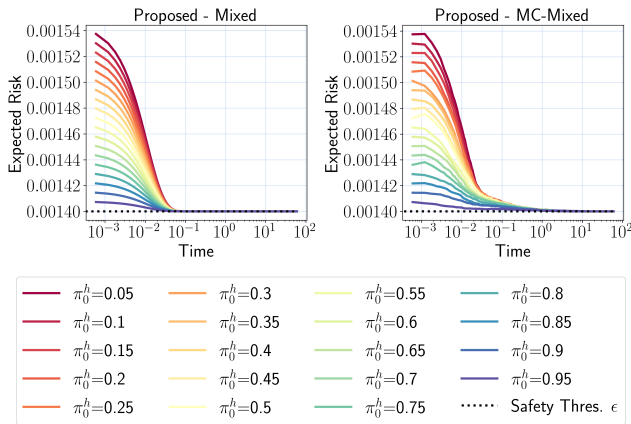


Figure 6: Robustness of the proposed methods controlling risks within a tolerable range against mixed HAs’ update rules (**Left**: ODE, **Right**: Monte Carlo) given different initial conditions in a Type B interaction ($\epsilon = 1.4e-3$, $w_r = w_b = w_s = \frac{1}{3}$ and $G = 1$).

Limitations and Future Work The main limitations of our work are as follows. First, we assume that the intentions of agents do not change within each episode, which is not always the case. We plan to consider intention changes within single interaction episodes in the future to further analyze the safety of interaction and improve our method. Second, the intention of agents are limited to either cooperate or defect. To extend the applicability of our method to even more general situations, we will consider other kinds of intentions, such as some more neutral intentions between the aggressive intention and conservative intention. Furthermore, it is natural that the environment changes over time and the rewards and risks may also change. Therefore, we will investigate how self-seeking humans will affect safety when we have time-varying interaction environments.

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